

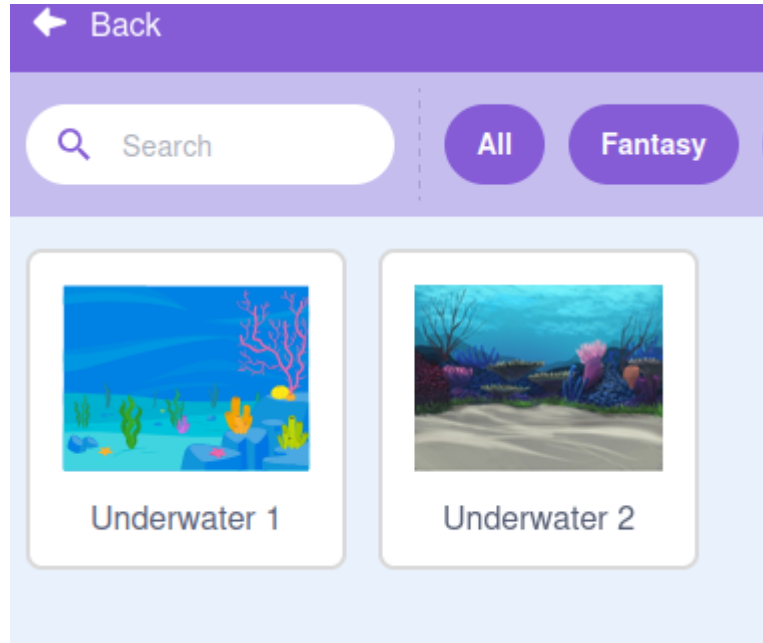
Scratch game from Scratch

Recap

- Variables
- Data types
- Operators
 - Logical
 - Arithmetic
 - Comparison
- IF, ELSE, ELIF
- Loops
 - For
 - While

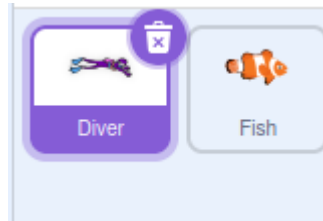
Scratch

Select your Stage/backdrop



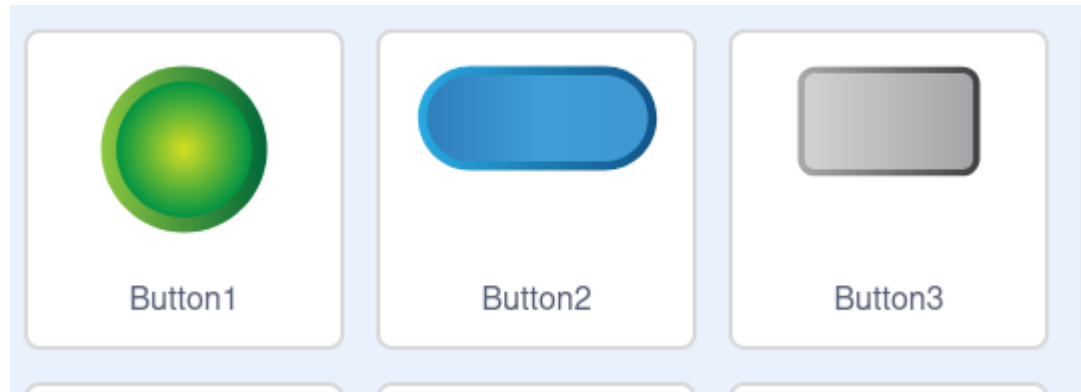
Sprites

Select two sprites:



Button

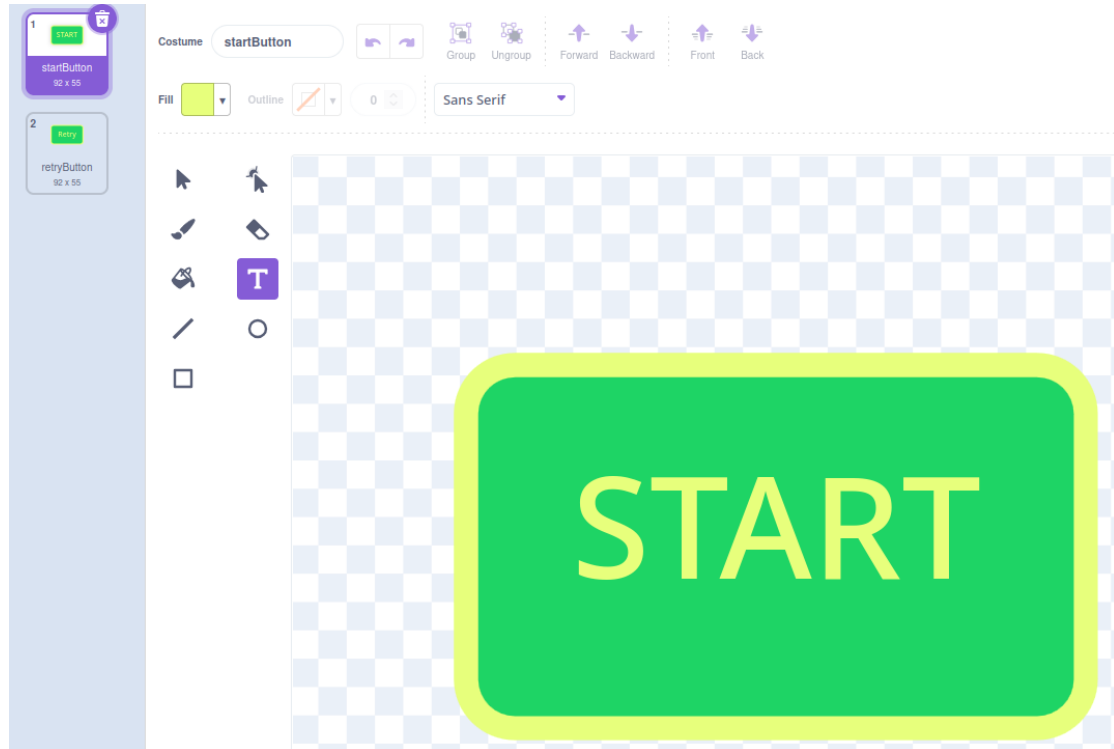
Select a button sprite



Edit button sprite

Edit button sprite to show costume1 – READY and costume2 – RETRY

Rename Costume:
startButton and retryButton

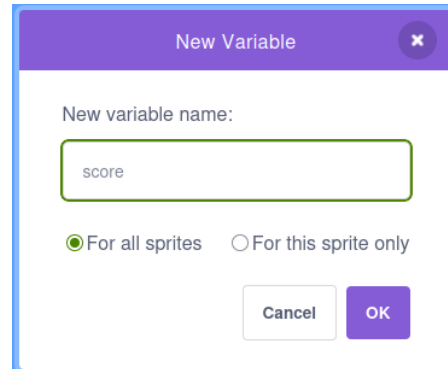


Variables

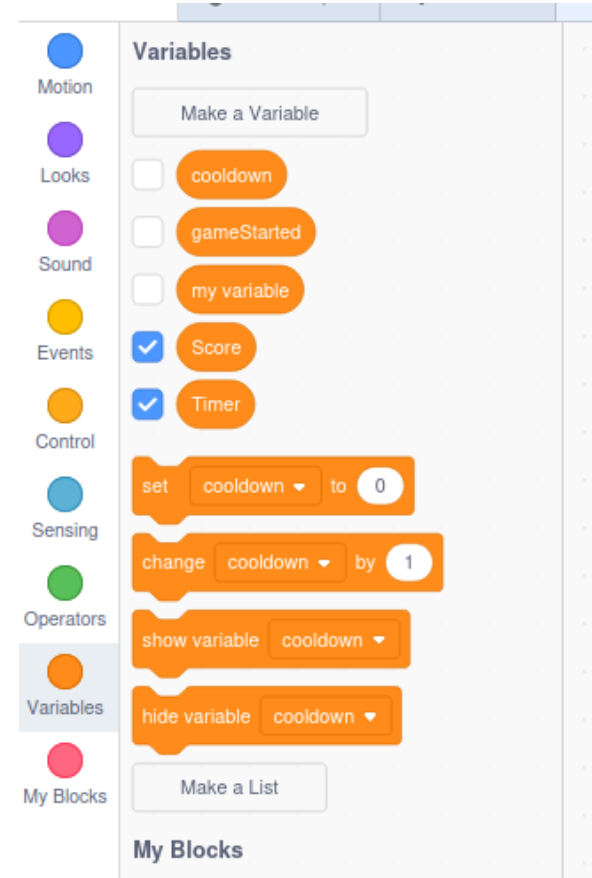
Set variables:

- cooldown
- gameStarted
- score
- timer

Make sure you set the variables `For all sprites`



The 'New Variable' dialog box is shown. It has a purple header with the title 'New Variable' and a close button. Below the header, there is a text input field labeled 'New variable name:' containing the word 'score'. Underneath the input field, there are two radio buttons: 'For all sprites' (which is selected) and 'For this sprite only'. At the bottom right, there are two buttons: 'Cancel' and 'OK'.



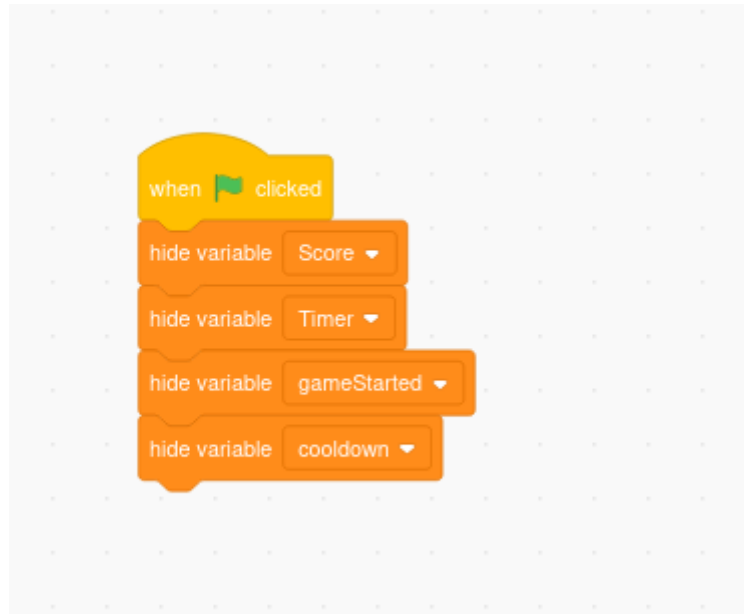
The 'Variables' panel in Scratch is shown. On the left is a category list with icons for Motion, Looks, Sound, Events, Control, Sensing, Operators, Variables (highlighted), and My Blocks. The main area is titled 'Variables' and contains a 'Make a Variable' button. Below this button is a list of variables: 'cooldown', 'gameStarted', 'my variable', 'Score', and 'Timer'. Each variable has a checkbox to its left; the checkboxes for 'Score' and 'Timer' are checked. Below the list, there are three variable blocks: 'set cooldown to 0', 'change cooldown by 1', and 'show variable cooldown'. At the bottom, there is a 'hide variable cooldown' block and a 'Make a List' button. The 'My Blocks' section is visible at the very bottom.

Backdrop

Start by hiding variables on

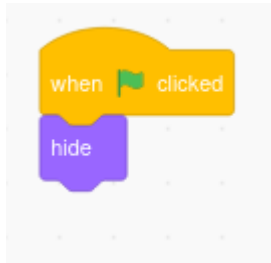


in your backdrop

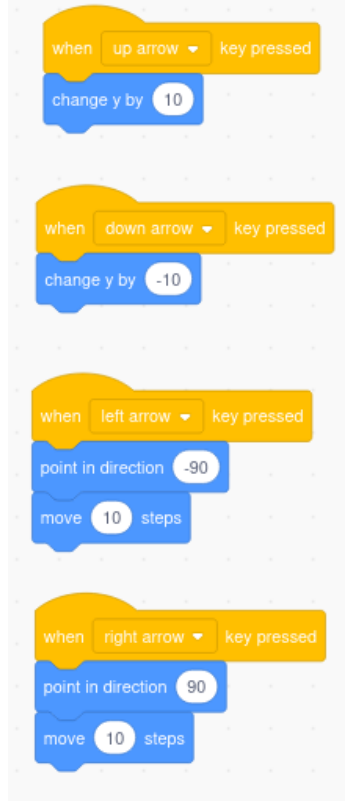


Your Character

Start by setting up your Character to hide when starting.

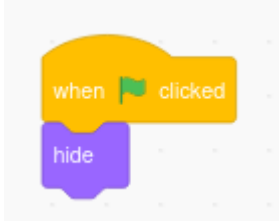


Set up character movements



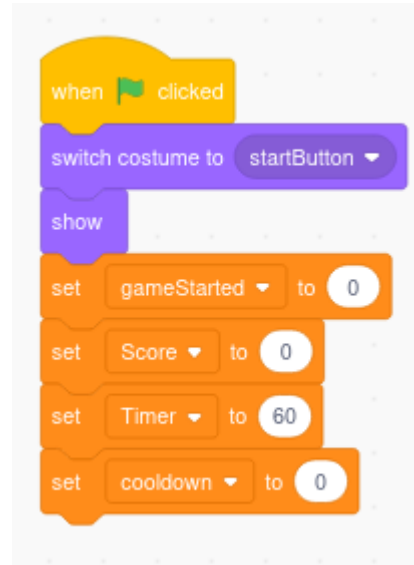
Set up NPC(Non Playable Character)

Start hidden



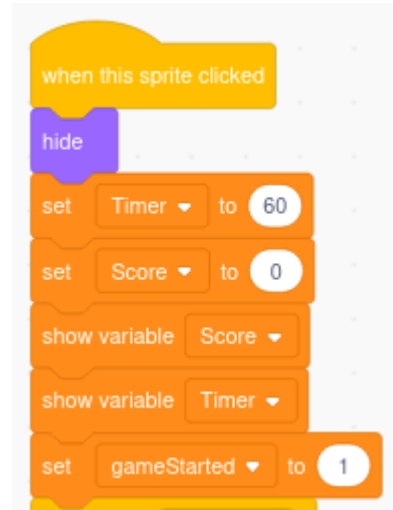
Button

The `Button` is the main function of the game that controls the everything, Start by setting the main event.



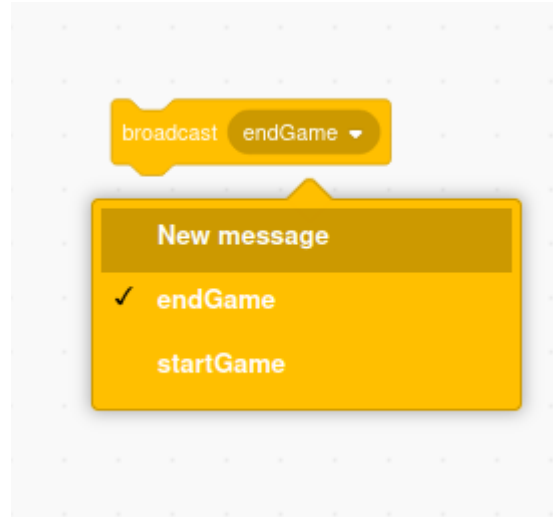
Click event

When we click the `Button` we want our game to start and `hide` the button and then set the variables we need for the game.



Broadcast

We need to let other items/sprites in the program aware the game has started so we do a `broadcast`. Create two `broadcast` messages endGame & startGame and add it to your last block.



Game Loop

This is the main game loop that will run and start the timer. This is the same as the while loop.

This could be written like:

```
1 while timer > 0:  
2   timer = timer - 1  
3   time.sleep(1)
```

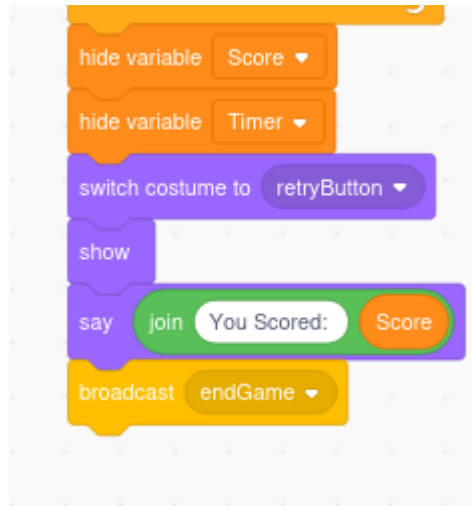


Or:

```
1 while timer > 0:  
2   timer -= 1  
3   time.sleep(1)
```

End of Game

At the end of the game loop we want to reset so we hide, score, timer and change the `button` costume to the `retryButton`, we then want to show the button and print the score and broadcast `endGame`.



NPC Logic

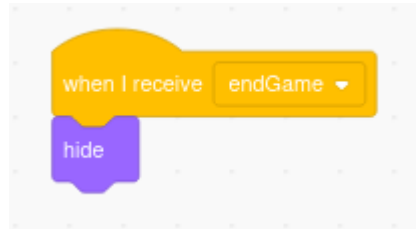
Now we have the game loop we can now work on the NPC Logic. We have already set it hidden, but now we want it to be visible when the game has started and to move around.

- Receive startGame broadcast from button
- Show
- Forever loop
- If gameStarted > 0
 - show
 - go to random position
 - repeat until gameStarted
 - wait 1 second
 - glide 1 second to random position
 - point towards Character
- Else
 - Hide
 - Stop



End NPC/Character

To finish of the NPC logic, we need to hide when game has ended. Do this for both NCP and Character



Character Logic

We need to follow similar logic to the NPC but the Character has a score and actions.

We will start with the game logic, When it receive the `startGame` show the sprite then run the loop to check if game has started, if not then hide.

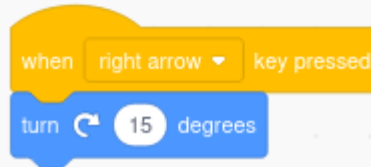
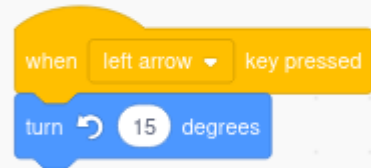
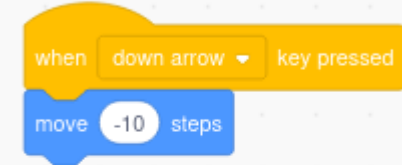
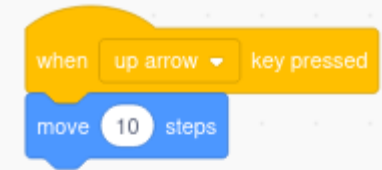


Character Actions

The character will have movement and an action for scoring.

Create events for Moving up/down and turning using the arrow, keys.

We will come back to this and fix it up for better game play later.

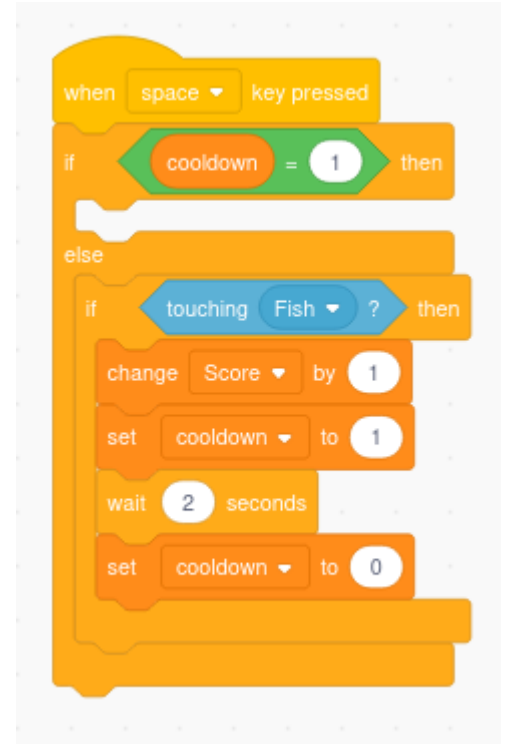


Scoring

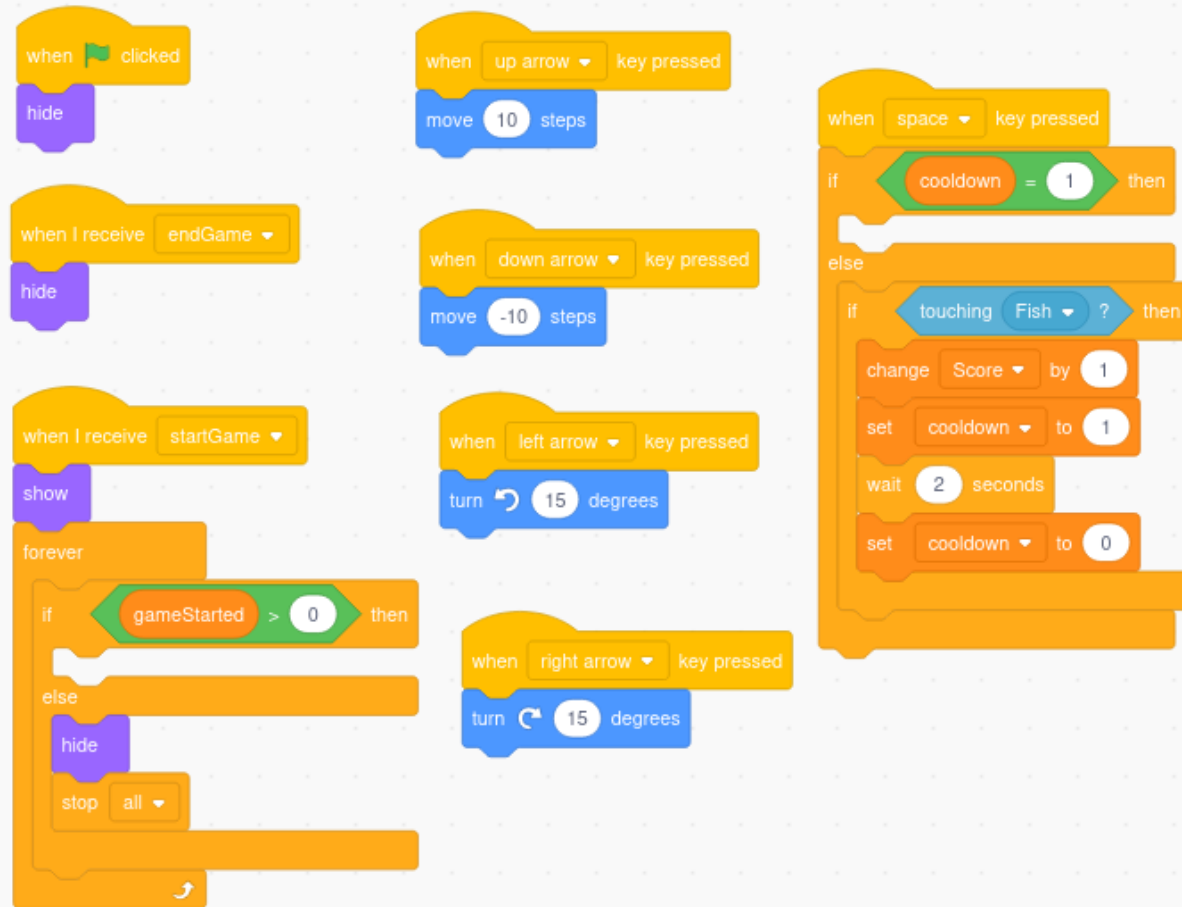
For the Character to score we need to be touching the NPC and press an extra button, in my example I chose `space`. Just so we can't spam space to score when we are touching the NPC we add a cooldown.

First we check if we are in a `cooldown`.

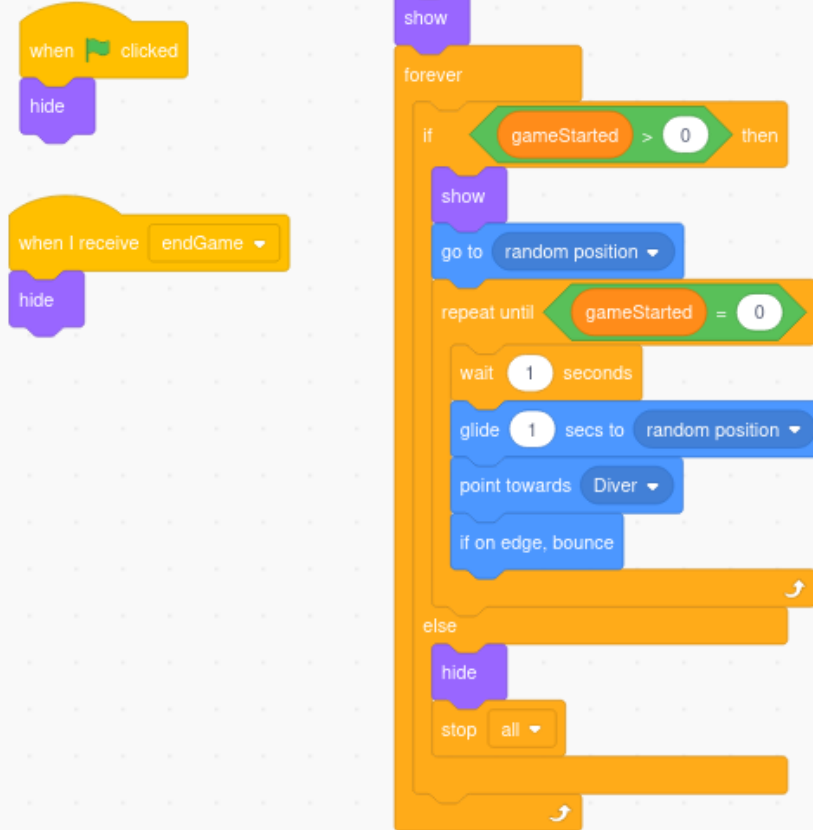
Then we check if we are touching the NPC(fish), then we change the score by 1 and set the `cooldown` to `1` and wait 2 seconds, before setting `cooldown` back to 0



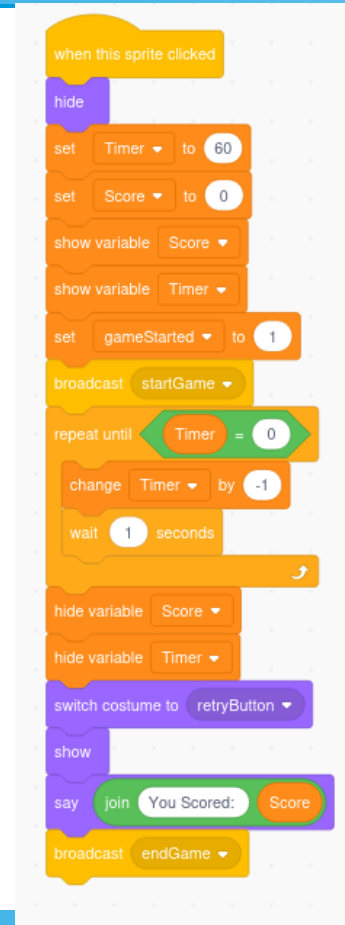
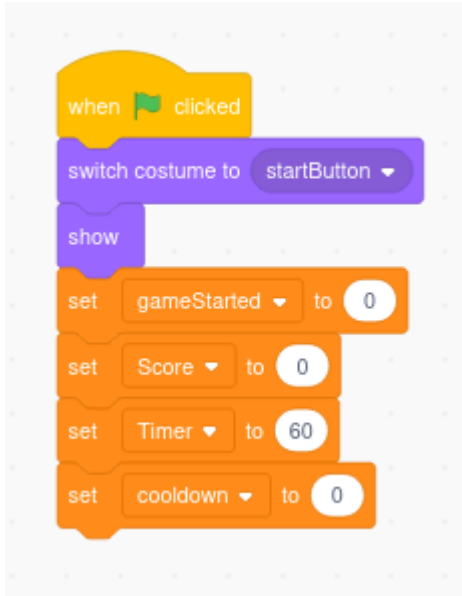
Recap - Character



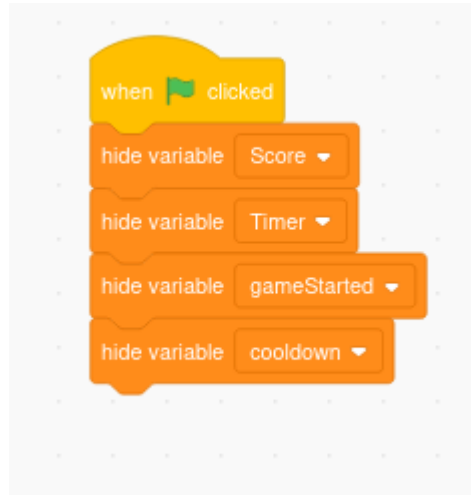
Recap - NPC



Recap Button



Recap - Backdrop



Play



Improvements

Try changing your Character movements to something like below and play and see if it feels better.



Movement Improvements

Try changing the left and right arrow press to change direction and move instead of rotate...

